

The Value of Ratings: Evidence from Their Introduction in Securities Markets

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We study the first-ever ratings for corporate securities. In 1909, John Moody published a book that partitioned the majority of listed railroad bonds into letter-graded ratings. These ratings had no regulatory implications and were largely explainable using publicly available information. We find that lower-than-market-implied ratings caused a rise in secondary market bond yields and that bonds that were rated experienced a substantial decline in their bid-ask spreads, consistent with reduced information asymmetries and improved liquidity. Our findings suggest that ratings can improve information transmission and highlight their potential value for the functioning of financial markets.

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I. Introduction

Bond markets are unique, not only for their economic importance but also for their unusually heavy reliance on ratings. Yet the centrality of ratings in bond markets constitutes something of a puzzle. In principle, bond ratings convey risk assessments to investors, thereby helping to solve problems related to asymmetric information. Yet in practice, it is unclear whether those ratings actually offer anything new to financial market participants, who would seem to have powerful economic incentives to obtain and process all information relevant to securities prices.

A vast empirical literature has established that ratings have significant impacts on a broad range of firm and market outcomes and has generally cast these effects as evidence that ratings are an important instrument for information provision.¹ Yet since the 1930s, credit ratings have become inseparably intertwined with explicit and implicit financial regulations, investment mandates, and bond covenants (White 2010), and it is these regulations and mandates that may explain the centrality of ratings in bond markets, as well as their effects. Even if they conveyed no information about credit risks, ratings could still have large impacts through their effect on the demand for securities by investors that operate within the purview of ratings-based regulations or mandates.² It is therefore not possible to determine empirically whether the impact of modern ratings is the product of information provision, by which we mean that their risk assessments conveyed via partitioning would affect market participants' expectations in the absence of mandates and regulations.³

To isolate the role of information provision in ratings, one has to turn to historical contexts prior to the introduction of rating-based regulations and investment mandates. In this paper, we analyze the effects of the first-ever securities ratings, introduced by John Moody in 1909. What turned out to be one of the most significant financial innovations in US history

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¹ See, e.g., Kaplan and Urwitz 1979; Holthausen and Leftwich 1986; Hand, Holthausen, and Leftwich 1992; Dichev and Piotroski 2001; Graham and Harvey 2001; Vassalou and Xing 2005; Kisgen 2006; Jorion and Zhang 2007; Sufi 2009; Ellul, Jotikasthira, and Lundblad 2011; Kisgen 2012; Aslan and Kumar 2015; and Almeida et al 2017.

² Any ratings change, even if completely exogenous, will lead investors to update the probability distribution over future ratings, distorting demand through future expected mandates and regulation, even if the change had no immediate regulatory implications. For example, a negative ratings watch, which has no regulatory implications, could increase the expected likelihood of falling below investment grade in the future and impact expected demand for the security.

³ This is distinct from whether ratings improve equilibrium information. A third-party evaluator might substantially impact markets using ratings and yet happen to be mistaken *ex post*, making prices less accurate.

was simply offered for sale in a printed volume in April of that year, as the product of a novel business venture that was not anticipated by investors and was not linked to any regulation or mandate. We find that Moody's ratings caused changes in bond yields and a reduction in information asymmetries, as reflected in a narrowing of bid-ask spreads. These results suggest an important role for ratings as tools for information provision, even in the high-stakes setting of financial markets.

The potential for a simple, discrete risk assessment to transmit information to investors is highlighted by another important feature of the origins of ratings. Moody had previously operated a business that produced "investor manuals" that presented much of the accounting data he later used to produce the ratings and had even published a volume that explained the evaluation of the credit risk of bonds (Moody 1906). His publication of ratings had significant market impacts above and beyond his earlier publications that conveyed much of the information on which the ratings were based. This suggests that what may have mattered was not only the level of information available but also the manner in which this information was presented to investors.

To identify the effect of Moody's 1909 ratings on market yields, we utilize a difference-in-differences design surrounding their publication, which compares the bond yields of firms whose ratings were likely to have been interpreted as a negative surprise to the yields of the bonds of other firms receiving the same rating but not as a negative surprise. We define negative surprises as cases where Moody's ratings were worse than investors' expectations, as implied by the risk assessments reflected in secondary market yields. In addition to estimating whether ratings surprises affected yields, this design also implicitly tests for convergence in yields within ratings levels. Our findings, however, are robust to the use of alternative empirical specifications in which we instead compare bonds with similar initial yields that did and did not receive negative surprises.

We find that negative surprises produced modest but appreciable changes in bond yields of about 16–25 basis points on an annualized basis, representing about a 2.9%–5.3% increase relative to average yields. We show that there is no evidence of differential trends prior to the release of ratings and that yields rose steadily afterward, consistent with a causal interpretation of our findings. We also show that a few months after Moody introduced his ratings, the financial press began to publish tables that compared the yields of different bonds that received the same rating and recommended trading based on the differences. This evidence suggests that investors quickly started to perceive similarly rated bonds to be of equivalent risk, consistent with surprises leading to convergence in yields within rating levels.

The relatively high-frequency nature of the yield response reduces concerns about confounding events. Although our estimated effects are

robust to the inclusion of a wide variety of controls, it is still possible that the firms that received negative surprises were different from the others along unobservable dimensions. If those firms happened to have differential exposure to a shock that coincided with the release of the ratings, that could potentially confound identification. To help address such concerns, we exploit within-firm variation by including firm fixed effects interacted with time trends and utilize differences in surprises among the bonds of the same company. Although the thinness of the trading of individual bonds adds noise to the surprises calculated in this way, the results with this specification are similar, suggesting that differences in firm characteristics are unlikely to account for our findings.

As with modern ratings (e.g., Benmelech 2017), we find that Moody's 1909 ratings were largely explainable from readily available information. His volume focused only on railroads, which were the dominant securities issuers of the time, and several competing firms published annual investor manuals containing financial information for those firms. These data were detailed enough to calculate the financial ratios that Moody used in compiling his ratings, and we show that his ratings are largely reproducible using these data. Yet the differential effects of negative ratings surprises on bond yields survive the inclusion of a wide variety of ratings determinants in our regressions, including the bonds' yields prior to the introduction of the ratings (our measure of investors' expectations). If ratings were largely predictable, how could they have conveyed information to market participants? We posit that the simplicity of a rating system with easy-to-interpret letter grades helped marginal investors become better informed.

Of course, ratings were not the only mechanism available to manage issues related to asymmetric information. The reputations of public companies in the early twentieth century were bolstered by the presence of elite investment bankers on their boards of directors, who monitored managers and represented the interests of securities holders (Frydman and Hilt 2017). If ratings helped to moderate problems related to information asymmetries, we would expect the bond yields of firms with elite bankers on their boards to see smaller effects. Consistent with this, we find that firms connected through their boards with top underwriting firms did not experience much of an increase in yields when they received a negative surprise. These heterogeneous impacts indicate that the effects of ratings may have been mediated by a firm's reputation, bolstering the interpretation that the introduction of ratings may have helped to reduce information asymmetries.

We then more directly explore the extent to which ratings affect the degree of information asymmetries among market participants. To the extent that ratings are able to resolve these asymmetries, we would expect to see a decline in market-based proxies such as bid-ask spreads. For bid-ask spreads to tighten, ratings must reduce the advantage of the most

informed traders (e.g., Glosten and Milgrom 1985). This may be the case if less-informed investors had higher costs of information acquisition and benefited differentially from the publication of ratings.

To assess how bid-ask spreads responded to the presence of ratings, we need a different empirical strategy than what we used for yields. Fortunately, Moody's volume did not rate all listed railroad bonds, and we can compare changes in the bid-ask spreads of rated bonds to those of railroad bonds that were not rated. The bonds Moody rated were, however, a selected group, with low initial yields and narrow bid-ask spreads on average. We address this selection problem by constructing an instrument based on Moody's rating procedure. Moody's volume rated all of the bonds of railroads that had at least some high-quality, low-yield bonds outstanding and were therefore of great interest to investors. The railroads that had no such bonds outstanding tended not to be rated. But due to their origins as amalgamations of many smaller carriers, the railroad systems of 1909 often had complex capital structures with large numbers of bonds outstanding, and sometimes railroads that had high-quality bonds also had some low-quality bonds, with higher yields and bid-ask spreads.⁴ These latter bonds were rated purely because they were part of the capital structure of a railroad that also had other, more important issues outstanding.

We therefore use the average yields of the other bonds issued by the same railroad as an instrument for whether a bond got rated. Results using this instrument in a two-stage least squares (2SLS) estimation show that the ratings resulted in substantially lower bid-ask spreads, consistent with more liquid and well-functioning financial markets. We also present suggestive evidence that ratings may have increased the trading of more illiquid securities by small investors, which is consistent with ratings helping marginal investors to become better informed.

Taken together, our results imply that Moody's letter-graded bond ratings constituted a valuable financial innovation that helped resolve problems related to asymmetric information. Institutional and retail investors quickly began to rely on them (Harold 1938), and their later incorporation into financial regulations resulted from their credibility. Despite the historical focus of our analysis, our paper offers insights relevant for modern financial markets. In the conclusion, we discuss the broader lessons that can be gleaned from our study.

The results of this paper advance an extensive literature seeking to understand the determinants and effects of credit ratings. Recent contributions

⁴ Railroad bonds were usually mortgage bonds; the bonds of the smaller railroads that were acquired by a larger system were backed by those railroads' specific railroad tracks, which could vary substantially in value. In many cases, the smaller railroads' bonds remained outstanding after the railroad was acquired by a larger system, meaning that the larger system's capital stock consisted of many different bond issues with collateral of varying quality.

to this line of research have obtained convincing identification by focusing on credit rating refinements that introduced more granularity in established ratings systems (Kliger and Sarig 2000; Tang 2009). Since the refinements can be treated as exogenous ratings changes, this empirical strategy isolates the impact of ratings from the impact of factors that influence ratings. These studies have interpreted the effects of ratings on prices, liquidity, and corporate outcomes as evidence that ratings convey information. However, this strategy cannot identify whether the effects of ratings are actually due to information provision as we have defined it, given the importance of ratings in financial regulations and investment mandates.⁵ By focusing on a historical environment in which there were no ratings-based regulations or investment mandates of any kind, our paper provides strong evidence that credit ratings can actually transmit information to market participants and reduce information asymmetries.

Utilizing a historical event to gain insights into the role of credit ratings naturally raises the question of external validity. It may be tempting to conclude that the information-based effects of ratings we identify in 1909 would be irrelevant today, given the improvements in financial disclosure, data availability, and information processing technologies that have emerged since then. However, it is important to note that the modern reliance on credit ratings by regulators and investors also affects information production in bond markets. Modern financial institutions other than credit rating agencies have limited incentives to invest resources in performing their own risk assessments, especially for highly rated securities (see, e.g., Johnston, Markov, and Ramnath 2009; Hanson and Sunderam 2013; Dang, Gorton, and Holmström 2020).⁶ Modern credit ratings may therefore continue to have significant information-based effects, although for somewhat different reasons than they did in 1909.

Our study also connects to a body of research studying how the institutional structure governing the production of modern ratings affects the quality of information production. In the wake of the Great Recession, modern credit rating agencies came under harsh criticism, and serious concerns were expressed about the quality of the ratings they produce, as well as the heavy reliance on ratings by both investors and financial

⁵ Any change in a rating to a level closer to (or further from) a threshold such as investment grade would change the current or future expected demand for a bond, even if it had no influence on investors' expectations of the likelihood of default. For evidence of rating-related regulation affecting investor demand, see, e.g., West 1973; Partnoy 1999; Moreau 2009; Kisgen and Strahan 2010; Lemmon and Roberts 2010; Bongaerts, Cremers, and Goetzmann 2012; Chernenko and Sunderam 2012; Flandreau and Sawatyniec 2013; White 2013; Bernstein 2017; Bao, O'Hara, and Zhou 2018; and Baghai, Becker, and Pitschner 2022.

⁶ Benmelech and Bergman (2018a, 2018b) and Johnston, Markov and Ramnath (2009) provide evidence consistent with the view that institutions other than credit rating agencies do not have a well-developed infrastructure for information gathering about bonds issued with high ratings.

regulators. A sizable literature shows that the issuer-pays business model that was adopted by ratings agencies beginning in the 1970s created perverse incentives that distorted the ratings given to many securities (Benmelech and Dlugosz 2009, 2010; Becker and Milbourn 2011; Bolton, Freixas, and Shapiro 2012; Griffin and Tang 2012; He, Qian, and Strahan 2012). The results of our paper suggest that when implemented simply as an evaluation of securities that were sold to investors, not issuers, and with no role in financial regulations, ratings were influential and likely beneficial for bond markets.

Our paper also speaks to a theoretical literature that has sought to understand the use of ratings or grades by information intermediaries. In these models, a third party in possession of private information will optimally choose to release only a coarse signal.⁷ Although these models help inform our analysis, none is entirely consistent with the settings for the 1909 securities ratings. Moody relied primarily on publicly available data to construct his ratings, did not charge railroads for rating them, likely had no reason to seek to influence railroad outcomes, and did not need to induce railroads to participate in his ratings system. The influence of Moody's innovation suggests that discrete ratings may have simplified the interpretation of data that were accessible but complex and difficult to evaluate, even in the context of a market in which participants had strong financial incentives to incorporate information into prices.⁸ Although we lack the data to test for these specific mechanisms, it is possible that by simplifying complex information, Moody's ratings may have helped small investors overcome complexity aversion (Puri 2022) or make better investment decisions in the face of a potentially overwhelming range of bonds and bond characteristics (e.g., Carpenter et al. 2021).

Finally, our paper also contributes to a significant literature analyzing the origins and effects of historical financial innovations in the United States.⁹ A number of related works have focused on early mercantile and securities ratings agencies (Chandler 1956; Sylla 2002; White 2010, 2013;

⁷ Prominent contributions to this literature have shown that a coarse signal maximizes revenue from fees charged to rated firms (Lizzeri 1999), maximizes outcomes of all graded students (Ostrovsky and Schwartz 2010), balances the value of the signal to rated firms and to investors (Goel and Thakor 2015), and expands participation in the rating system (Harbaugh and Rasmusen 2018), often in the context of a cheap talk game (e.g., Crawford and Sobel 1982; Martel et al. 2022).

⁸ These results relate to Luca and Smith (2013), who show in the context of college ratings that the order in which these ratings are listed has very different effects on college applications, even when the data and methodology to construct these ratings are provided.

⁹ Noteworthy examples include securities markets (Rousseau and Sylla 2005; Atack and Neal 2009), bank clearinghouses (Gorton 1985; Tallman and Moen 2012; Jaremski 2018), commercial paper markets (James 1993; Calomiris, Himmelberg, and Wachtel 1995), shadow banks (Gorton and Metrick 2010; Frydman Hilt, and Zhou 2015; Rockoff 2018), central banks (Sylla 2010; Bordo and Roberds 2013), and collateralized banknotes (Rolnick and Weber 1983; Gorton 1996; Jaremski 2010).

Flandreau and Mesevage 2013; Bernstein 2017; Pénét 2019; Carruthers 2022) and the predictive power of historical securities ratings (Harold 1938; Hickman 1958; Hempel 1971; Flandreau, Gaillard, and Packer 2011). Yet surprisingly, despite their transformative effects on bond markets, Moody's first-ever securities ratings have not been the focus of much empirical analysis. One notable exception is Wilson (2011), who analyzes the determinants of Moody's 1909 ratings. Our paper focuses instead on empirically estimating the effects that those ratings had and shows that ratings can improve the functioning of bond markets and market access.

II. Origins of First Letter-Graded Security Ratings

A. Early Twentieth-Century Bond Markets

Well before the introduction of credit ratings, the corporate bond market in the United States grew to become quite extensive. Figure 1 presents data on the market's evolution from 1880 to 1910. Figure 1A shows that total outstanding corporate debt securities grew from \$2 billion to \$15 billion over this period (nominal gross domestic product [GDP] in 1910 was about \$30 billion). Railroads dominated the market, although after 1900 the volume of bond issues of utilities and industrials grew rapidly.

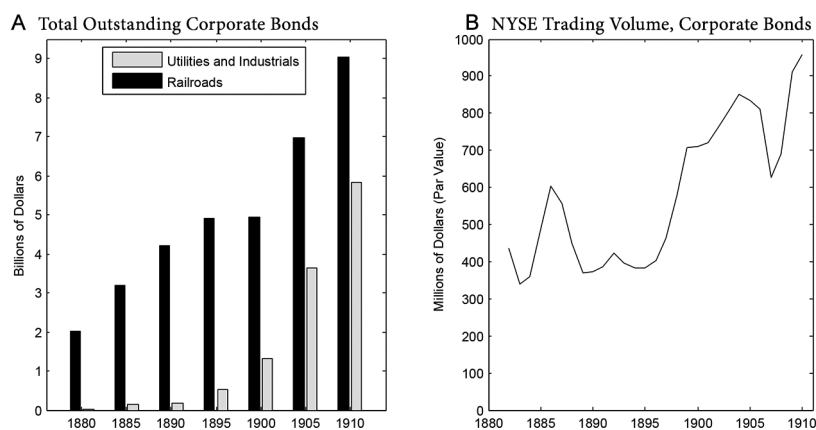


FIG. 1.—Evolution of US corporate bond markets, 1880–1910. A, The total value of outstanding corporate bonds, at 5-year intervals, by sector. B, A 3-year trailing moving average of total NYSE bond transactions, expressed in millions of dollars of par value. Data for utilities and industrials were compiled from the *Commercial Financial Chronicle's* “Investors’ Supplement,” for years prior to 1900, and for later years, from Hickman (1952); railroad data are from Hickman (1952). For years after 1900, trading in “railroads and miscellaneous bonds” was recorded from the *New York Times* “Annual Financial Review”; data for earlier years was obtained from Stedman (1905).

Whereas today bonds are primarily traded over the counter, during this period they were generally listed on the New York Stock Exchange (NYSE) and traded within a special section of the exchange floor. Figure 1*B* shows that the volume of trading in bonds on the NYSE grew substantially but not as rapidly as the volume of outstanding bond issues. Retail investors accounted for a substantial share of the trades executed on the exchange (Meeker 1930, 260), and single-lot trades, which were likely to have been ordered by retail investors, accounted for around 20% of trades.¹⁰ Institutional investors, such as insurance companies, also traded through brokers on the exchange floor.¹¹ In general, trading was concentrated on a relatively small number of well-known railroad bonds; the vast majority of NYSE-listed bonds traded only infrequently.

As the bond market expanded, information resources for investors proliferated as well. By the turn of the twentieth century, several competing firms produced annual volumes of financial data on major corporations, which included Henry Varnum Poor's *Poor's Manual of Railroads*, Standard Statistics Service's *Manual of Statistics*, and John Moody's *Moody's Manual*. However, these investor manuals did not provide much assistance with interpreting the information they contained. They simply presented firms' financial statements and offered little additional commentary or analysis. The volumes were sold at a relatively high price and marketed to sophisticated investors, most likely the bond departments of private banks, insurance companies, and commercial and savings banks, which possessed the requisite knowledge of accounting and finance to analyze the information presented.¹²

Perhaps as a consequence of the availability of these information resources, coupled with the widespread adoption of communications technologies such as the telephone and telegraph, early-twentieth-century bond prices generally reflected information on credit risks. For example, during the years 1899–1919, the correlation between the prices of high-quality bonds and corporate defaults in the following year was -18.3% . This is quite similar to the modern era; for the years 1996–2010, the correlation was -18.2% .¹³

¹⁰ The face value of bonds at the time was typically \$1,000, which was more than twice nominal GDP per capita at the time, so retail investors were wealthy individuals. We calculate the fraction of trades by size in our intraday trading data below; see table A.5 (tables A.1–C.1 are in the online appendix).

¹¹ There was also an over-the-counter market in which institutional investors transacted directly with one another or with a broker over the telephone; this market grew significantly in importance during the 1920s. See Meeker (1930).

¹² The list price for most of these volumes was around \$12 in 1909. Adjusting for inflation using the Consumer Price Index, this is equivalent to more than \$350 in today's money.

¹³ These calculations were made using bond price data from Homer and Sylla (1996) and corporate default data from Giesecke et al. (2014).

B. *Moody's Innovation and Its Reception*

In the wake of the Panic of 1907, John Moody was forced to sell his Moody Manual Company, and these manuals continued to be produced under his name by a competitor. To reenter the market with a differentiated product, he established a new business that was focused on providing investors with analysis rather than just financial data. His new Analyses Publishing Company produced its first annual volume, *Moody's Analyses of Railroad Investments*, in April 1909. The volume provided data summarizing the financial statements of every major railroad and applied a letter-graded rating system to categorize the credit risk of their bonds, nearly 1,300 in total—quite akin to modern ratings. Importantly, the publication of the manual and, especially, the ratings it contained were unanticipated by market participants.

The 1909 volume was quite different from a typical investor manual, in that it included long chapters explaining railroad financial statements and how they should be evaluated. And whereas typical investor manuals included large amounts of advertising, with ad sales producing a significant source of revenue (Chandler 1956), Moody's volume of ratings contained no advertisements, which eliminated any possibility that ad spending might influence its content.

At the time, commercial credit reporting agencies such as R. G. Dun & Co. and Bradstreet's offered letter-graded ratings of the creditworthiness of individuals and firms, and these mercantile ratings likely inspired Moody. His innovation was to develop a quantitative system to rate the specific bond issues of railroads, in a volume that included clear descriptions of the specific data and methods used. Moody's ratings scale could take 11 different values ranging from Aaa to E, which offered investors a simple summary measure of the quality of a bond, in a hierarchy that was easily understood.

The methods Moody employed in his evaluation of the bonds were not new. Many published guides for investors available at the time noted that the safety of railroad bonds could be evaluated by comparing their interest obligations to the railroad's earnings, which, as we will show below, is essentially what Moody did. In fact, Moody himself published a volume in 1906 that argued for doing this and described the approach he would later adopt in his volume of ratings (Moody 1906). What was new was that Moody painstakingly calculated these statistics for most railroads and developed discrete thresholds characterizing different levels of safety—his ratings. Other sources had suggested rules of thumb for what constituted a safe bond or a risky bond, but no investor guide had ever developed a consistent rating system, much less actually applied it systematically.¹⁴

¹⁴ Nelson (1907) is an example of the former and stated that “a first-class bond investment necessitates that a road should earn double its fixed charges.” Another example is Hall (1906, 32), who stated that “from 60 to 65 percent of the profits should pay all fixed charges, that is to

For each railroad in the manual, Moody presented 10 years of earnings and expense data and compared their values (scaled by mileage) with those of other railroads in the same region. Moody then presented simple tables that listed the railroads' bonds and offered a rating based primarily on its "factor of safety"—the share of earnings remaining after interest on the bond and all bonds senior to it had been paid.¹⁵ Summary descriptions of Moody's ratings are presented in table 1. The overwhelming majority of the bonds rated in the volume received ratings of Aaa, Aa, or A, reflecting the relatively high quality of the bonds of railroads, which were then the "blue chip" companies.

Similar to other investor manuals, Moody advertised his volume in the financial press. But he also appealed directly to retail investors, taking out ads in the *New York Times* that mentioned prominent railroads and touted the volume's analysis of their securities. The ratings offered a convenient and comprehensive guide to the quality of different railroad bonds that was independent from the securities dealers who marketed bonds. As many commentators noted, this was particularly valuable to small investors (e.g., Johnson 1909).

Moody's 1909 volume received high praise in the press for the care with which its statistics were calculated.¹⁶ Some commentary about the volume also noted that the ratings were based on publicly available information and that the value of Moody's system was that it could help investors evaluate that information.¹⁷ Other commentary characterized the ratings as "merely opinions," which was an effective way to minimize potential liability, but noted that they "have the merit of being presented along with the facts that gave rise to them."¹⁸

Although we lack detailed sales data, we know the volume sold well and that major public libraries and university libraries acquired it. Investment advisors' articles in the financial press, such as in *The Ticker*, commonly recommended that investors consult Moody's volume in response

say, taxes and interest on the funded debt. If 80 percent is required, an investor should take advice as to the propriety of selling his bonds and going into some other security."

¹⁵ See fig. A.5 (figs. A.1–B.1 are in the online appendix) for an example.

¹⁶ For example, the *New York Times* praised its "complete original analyses of all the leading railroad systems" (April 30, 1909); the *Railroad Age Gazette* stated that "the work of calculation has been done in a careful and scholarly way" (April 30, 1909); and *American Review of Reviews* called it "not a manual, but a commentary, ingenious, painstaking, and authoritative."

¹⁷ "The book tries to give for each railway and for each bond the statement which would be asked from the statistician of a good private banking house by the partners, when the railway or the bond was under consideration. With the exception of certain kinds of transitory and confidential information which the statistician would probably possess, the record in this book is as complete as need be, and the book is far better adapted for the use of the intelligent private investor than is any railway manual that has come to our attention" (*Railroad Age Gazette*, April 30, 1909).

¹⁸ *American Review of Reviews* (1909), vol. 39, 757.

TABLE 1
MOODY'S 1909 RATINGS

Rating	Description	Mean			
		Factor of Safety (%)	Income Per Mile (000s)	Seniority Rank (1 = highest)	Yield to Maturity (%)
Aaa	Highest class, value is not affected by any normal changes in earnings capacity of railroad itself	81.25	5.79	6.79	4.29
Aa	While high-grade, slightly inferior to those having first rating in security or in salability	74.11	3.28	8.64	4.24
A	Although high-grade, affected, to partial degree, by changing earning power	67.82	3.16	10.83	4.68
Baa	Generally good but have speculative tinge, good but second-grade issues	50.70	2.67	20.39	4.64
Ba	Make moderately favorable showing and regarded as well secured but affected by changing earning power	56.55	1.34	17.17	5.30
B	More susceptible to fluctuations and regarded as more speculative in position	43.90	1.38	13.22	5.31
Caa	Almost directly responsive to changes in earning power and have not had benefit of available income equal to more than double the interest	45.00	1.53	27.00	5.18
Ca	Approach more strongly to field of speculative issues with but moderate security	31.04	.97	16.80	7.92
C	Show but a slight margin in surplus above amount required for their interest and which are not well secured	20.00	1.17	16.58	6.88
D	Of doubtful character and almost purely speculative value				
E	Defaulted issues, awaiting the results of reorganizations				

NOTE.—Authors' calculations from data presented in *Moody's Analyses of Railroads Investments*, 1909, and from weekly bond prices reported in the *New York Times*. Average (mean) statistics are computed for each bond-week observation used in the primary regression analysis shown in table 4. "Factor of safety" is the percent of earnings available after interest on the issue (and other issues with equal seniority) has been paid. Income per mile is average earnings available for interest, per mile. Seniority rank is the ranking by Moody of the bond among all issues from the same railroad, with 1 being the most senior bond. The yield to maturity is the average value calculated from closing prices as reported in the *New York Times* over our sample.

to questions related to valuing securities. The volume was also published in London, where insurance companies, which often held American railroad securities, reported that it was quite valuable.¹⁹

¹⁹ The *Insurance Record* of December 3, 1909, noted that "[insurance companies] expecting a yield of at least 4 per cent. on foreign investments, would probably in future be disposed to

Moody's ratings quickly became influential. In contrast to his ads for the 1909 volume, his advertisements for later editions led with the question, "How Are Your Bonds Rated?"; this suggests that the potential value of bond ratings had become well understood among investors.²⁰ In a sign of the success of the concept, in 1914, Moody expanded his ratings to include public utility and industrial bonds. In addition, Moody's competitors also began to publish ratings in their volumes. The first was Poor's, which published its first ratings in 1916, then Standard Statistics followed in 1922, and finally Fitch began to publish ratings in 1923. Each of these firms rated securities according to a letter-graded scale similar to that of Moody's, and each followed Moody in selling the ratings to investors rather than having issuers pay for the ratings, as is done today.

The success of ratings also contributed to their use by financial regulators. This process began in the fall of 1931, when the Office of the Comptroller of the Currency held that national banks could hold highly rated bonds on their balance sheets at their book values, whereas lower-rated bonds had to be marked down to their market values.²¹ Over subsequent decades, ratings were incorporated into a wide variety of financial regulations, including rules governing money market mutual fund investments, the capital requirements of insurance companies, the investment criteria of pension funds, the investments of savings and loan associations, and computations of net capital for broker-dealers (Langohr and Langohr 2009; White 2010).²² It is precisely this reliance on credit ratings in financial markets that makes it infeasible to disentangle the effects of information embedded in ratings from changes in investors' demand due to ratings-based regulations and mandates in modern settings.

C. *Empirical Predictions*

We analyze two potential consequences of the introduction of ratings. The first focuses on the content of the ratings and its effect on bond yields. As noted above, prior to the emergence of ratings, investors had access to detailed financial data on all railroads; contemporary commentators noted that bond yields were influenced by reasonably well-informed assessments of their risks.²³

invest in a somewhat lower, but nonetheless well-secured, class of American railway bonds, and the statistics in Mr. Moody's book were of great assistance in forming a fair estimate of their security" (575).

²⁰ *Wall Street Journal*, March 8, 1912, 1.

²¹ Flandreau and Sawatyniec (2013) argue, however, that US courts began to rely on the judgments of ratings agencies in the 1920s, giving them a form of legal authority at an earlier date.

²² Credit ratings are now ubiquitous, and their role in the structure of the market goes well beyond regulations. For example, they are typically incorporated to the credit annexes of over-the-counter derivatives contracts.

²³ For example, Pratt (1912) noted that "the prevailing price of railroad bonds bears 4 per cent interest, and if of undoubted standing they command a premium."

We therefore focus on cases where ratings conveyed new information relative to investor expectations, as embodied in preratings market yields, and test for yield responses to the introduction of the ratings.

There is evidence that investors may have used Moody's ratings to guide their bond trading in ways that are consistent with this hypothesis. Beginning in November 1909, the popular investing magazine *The Ticker* began to publish a bond "Buyer's Guide" that compared the yields of different bonds with the same rating from Moody.²⁴ This suggests that investors may have accepted Moody's ratings as a measure of risk that they could compare with yields and that lower-yielding bonds of a given rating may have been regarded as having unattractive prices.

Our second test focuses on the effects of the ratings on quoted bid-ask spreads for bonds. As in any financial market, it is likely that problems related to asymmetric information reduced liquidity, as reflected in the bid-ask spreads quoted by dealers. Trading in bonds on the NYSE was facilitated by dealers known as specialists, who would make a market in particular issues. Investors wishing to purchase or sell bonds would place their order with an NYSE-member firm, which would telephone the order to the NYSE floor, where one of the firm's brokers would receive it and go to the specialist who handled trading in that issue to try to negotiate a sale.

The dealers on the NYSE floor would have had an informational advantage over relatively uninformed investors, but on the other hand, there were also investors who possessed private information about market conditions or railroads who had an informational advantage over dealers. Dealers knew that trades with the former would likely be profitable, but trades with the latter would likely be unprofitable. The fundamental problem faced by the dealers was they could not distinguish the two within the order flow they received.²⁵ This likely widened quoted bid-ask spreads, an implication that is largely consistent with modern theoretical underpinnings of drivers of the bid-ask spread (e.g., Glosten and Milgrom 1985). The introduction of securities ratings may have improved liquidity by changing the expected value of the informational disadvantage dealers faced with regard to their counterparties trading on the exchange. If the ratings increased participation in the market by small investors, who were likely to have been less informed than the dealers, or if they improved the dealers'

²⁴ November 1909 issue, 16. The table grouped all recently traded bonds by their rating levels and was described as presenting data on "the relative cheapness of principal railroad issues." Similar tables were printed in each subsequent issue over the following year. An example is included in the appendix as fig. A.1.

²⁵ Figure A.7 illustrates this issue; dealers transacted with brokers on the exchange floor, and the brokers executed orders sent to them by investors. But the dealers could not know whether particular investors placing orders were uninformed or possessed an informational advantage.

position relative to investors with private information, then quoted bid-ask spreads would have fallen.

Our analyses of the effect of ratings surprises on market yields and of the effect of receiving a rating on quoted bid-ask spreads both constitute tests of information provision by ratings. In the absence of any ratings-based financial regulations or investment mandates, any impact on yields will be an outcome of the ratings shaping investors' expectations of the riskiness of bonds, and any impact on bid-ask spreads will be the result of the ratings influencing the perceived informational advantage of better-informed investors of the market.

III. Data and Descriptive Statistics

Our analysis is based on novel, hand-collected data from various sources. In this section, we briefly describe our main sources and variables.

A. Bond Transactions

To analyze the impact of securities ratings on bond yields, we collect weekly closing prices for all railroad bonds traded on the NYSE, as reported on the Monday edition of the *New York Times*.²⁶ Figure A.2 presents a partial example of 1 week's worth of transactions. We collect these data for a total of 2 years, centered on the date when ratings were introduced. We restrict the data to the railroad bonds for which we observe at least one traded price before and after the introduction of ratings, resulting in an unbalanced panel comprised of 562 bonds corresponding to 67 different railroads. Table A.1 presents simple summary statistics for these data. About 90% of the transactions were for bonds rated by Moody (resulting in 454 bonds for 44 firms in our analysis that looks at surprises for rated bonds), corroborating our view that his 1909 volume assigned a credit rating for a large fraction of the railroad systems in the country.

B. Market Microstructure

In 1909, a relatively large number of bonds were listed on the NYSE. Each day at 11 a.m., the exchange printed and distributed quotations sheets that included all bid and ask quotations for listed bonds. Unlike the transactions data, this information was, to the best of our knowledge, only published in the press at a low frequency, rendering these sources unsuitable for our study. Instead, we access the original quotation sheets at the archives of

²⁶ The fact that bonds were listed on the exchange enables us to observe prices for actual transactions. For a discussion of the later shift of bond trading off the exchange, see Biais and Green (2005).

the NYSE. See figure A.3 for a partial example of these bid and ask quotations. We digitized these data at a weekly frequency for the 12 Wednesdays before and after the introduction of the ratings. We compute bid-ask spreads with these bonds, which will serve as our principal measure of liquidity, and restrict our data to the 346 bonds for which we observe at least one nonmissing spread in the 12-week period before and after the publication of Moody's ratings in April 1909.²⁷

Summary statistics for these data are presented in panel A of table A.3. About 94% of the quoted bond spreads in our sample were for bonds rated by Moody. There was a clear difference in liquidity and risk between rated and nonrated bonds. The mean bid-ask spread of the rated bonds was 1.1%, whereas it was 2.9% for the bonds that were not rated. In addition, the yields on the rated bonds were 4.3% on average, whereas they were 6.2% for the bonds that were not rated. Yet these differences in means mask substantial variation in the distribution of yields and spreads by rating status. For our analysis, it is important to note that there is substantial overlap in these distributions, with a high fraction of rated bonds having yields and spreads similar to those of nonrated bonds. For example, the 25th percentile of unrated bonds had bid-ask spreads and yields of 69 bps and 4.3% over the entire sample period, respectively, which are close to the median for rated bonds.

C. Ratings and Director Information

We utilize Moody's 1909 volume to collect the assigned rating, and all the information pertinent to how these ratings were assigned, at the bond level. We also collect the names of the directors of all railroads in the sample from the 1909 edition of *Moody's Manual*, the investor manual formerly published by Moody. As emphasized by Frydman and Hilt (2017), board interlocks between railroad firms and main financial intermediaries were common and of importance for easing financial constraints in the early twentieth century. In our analysis, we study how these relationships affected the impact of the introduction of credit ratings, by matching the names of railroad directors to lists of financiers. Following Frydman and Hilt (2017), we use bond underwriting data from the 1913 edition of the *Fitch Bond Book* to determine the financial institutions that were the top underwriters at the turn of the century and then match the directors and partners of those financial firms to the boards of railroads.²⁸ Most investment banks were partnerships and were members of the NYSE; we

²⁷ Specifically, we estimate the spread as $[(ask - bid) \times 2] / (ask + bid)$, and it is therefore expressed as a percentage.

²⁸ Specifically, we collect the names of the lead underwriters (where available) for all outstanding debt issues of NYSE-listed corporations and rank underwriters by volume in order to identify the top 10 firms. See Frydman and Hilt (2017).

obtain the names of their partners in 1909 from the NYSE Directory. For commercial banks and trust companies, director names were obtained from the 1909 edition of *Rand McNally Bankers' Directory*.

IV. Determinants of Ratings

We start by presenting simple summary statistics to illustrate the key determinants of Moody's ratings. The first two columns of table 1 present the ratings and their descriptions for all railroad bonds included in Moody's 1909 volume. Railroad bonds were considered to be relatively safe, with the vast majority of bonds being rated A or higher. The assigned ratings do appear to have been primarily a reflection of a bond's expected risk. While Aaa bonds were stated to be "not affected by any normal changes in the earnings capacity of the railroad," lower-rated bonds, like those rated B, were "[m]ore susceptible to fluctuations." This is consistent with simple summary statistics that we construct utilizing information for the securities that are included in our analysis, presented in the last four columns of the table. Better-rated bonds tended to have a higher factor of safety (average percentage of earnings available after paying interest over the previous decade) and income per mile. They were also more senior and had a lower yield to maturity.

Not only were the ratings correlated with many simple measures of risk, but the ratings tables in Moody's 1909 volume indicate that some of these metrics, such as the factor of safety, were used in determining the ratings (e.g., fig. A.5). Moody based his assessment on many of these metrics using long-run data spanning primarily the decade prior to the issuance of the ratings volume (see fig. A.6). In table 2, we analyze whether simple statistics help explain the observed variation in railroad ratings.

We start by converting bond-level ratings into mean ordinal rankings across all issues within a given firm, and then we take the average across all issues to estimate an issuer-specific rating. Similarly, we average other bond-level characteristics. Weighting to match the sample in our primary regression analysis, we find that the absolute value of pairwise correlations between ratings and average factor of safety, preratings bond yields, and average income per mile, are 82%, 56%, and 46%, respectively. These are incredibly high pairwise correlations for simple linear measures of each of these factors, which are primarily based on averages of multiyear lagged data originally reported in firms' annual financial statements that were then disseminated in investor manuals.²⁹ Combinations of these factors

²⁹ In unreported analysis, we use two competing investor manuals to reconstruct measures of factor of safety and income per mile for the 10-year period ending on 1909. We find correlations ranging from 0.8 to 0.98 between these alternative sources and the data listed by Moody's in 1909; the differences are primarily due to variation in the number of years' data reported. This suggests that the data on which Moody based his ratings were generally known, or at least knowable.

TABLE 2
RATINGS: PAIRWISE CORRELATIONS WITH FIRM CHARACTERISTICS

Correlations (firm level)	Rating (Aaa = 1)	Factor of Safety	Income per Mile	Interest per Mile	Duration	Bid-Ask Spread	Preperiod YTM (mean)
Rating	1						
Factor of safety	-.82	1					
Income per mile	-.46	.39	1				
Interest per mile	-.08	-.02	.60	1			
Duration	.13	-.11	.03	.44	1		
Bid-ask spread	.30	-.19	-.17	-.09	.05	1	
Preperiod YTM	.56	-.55	-.43	-.23	-.15	.38	1

NOTE.—Pair-wise correlations between firm-level characteristics that may be correlated with Moody's 1909 securities ratings for railroad companies. "Rating" is a simple ordinal ranking of a bond's rating in 1909, with the lowest risk group (Aaa) denoted as 1 and values increasing in increments of 1 from there (e.g., Aa = 2, A = 3, . . .). We calculate the median over all bond-week observations in our sample per firm to obtain a firm-level measure. "Factor of safety" is the percent of earnings available after interest on the issue (and other issues with equal seniority) has been paid. "Average income" is the average earnings available for interest, per mile. "Interest per mile" is the average interest owed per mile of railroad line. "Duration" is the bond's duration given its coupon rate and maturity. "Bid-ask spread" is the average difference in our sample pre-ratings period between the bid and ask prices from the same day at 11 a.m., when the exchange printed and distributed quotations sheets that included all bid and ask quotations for listed bonds. "Preperiod YTM" is the average yield to maturity calculated from closing prices as reported in the *New York Times* during the months prior to the publication of the ratings volume. Unless otherwise stated, we map bond characteristics to firms by computing the mean value of that variable for the firm across all bond-week observations in our main sample used for our analysis in table 4.

easily explain more than 80% of the total variation in the firm-level ratings within simple additively separable linear models, leaving limited room for factors other than public information in explaining the vast majority of the variation. That the first-ever ratings largely relied on observable firm characteristics may not be shocking, though, since John Moody tasked himself with rating almost 1,300 bonds. Yet credit ratings today, though much transformed, are also highly predictable, with linear combinations of a small set of firm characteristics (Benmelech 2017).

The especially predictive nature of prerating yields also motivates our empirical design. Bond yields can be thought of providing a plausible market-based proxy for perceived risk—all else equal, riskier bonds should have higher yields. A priori, it is unclear whether ratings should have any influence on markets above and beyond the information already contained in yields. To assess this possibility, we first study deviations between Moody's ratings assessments from preexisting market expectations.

V. Ratings and Bond Yields

A. *Empirical Methods and Predictions*

We study whether Moody's ratings had any incremental effects beyond what the information already contained in bonds' yields. In order to test for this, we focus on cases in which the ratings may have surprised market participants, relative to what their expectations on bond risk were prior to the ratings' existence. We proxy their expectations by the bond yields preratings. We define surprises as cases in which the rating assigned by Moody to a railroad was different from the ratings assigned to other railroads whose bonds had similar yields. This market-based approach enables us to compute a measure of the surprise content of the ratings that is independent of the specific algorithm that Moody may have used to calculate credit ratings.³⁰

To construct a measure of the surprises at the firm level, we first compute the mean yield to maturity of each railroad's bonds preratings. We then classify the railroads into quartiles of the distribution of mean yields.³¹ Table 3 presents the resulting yield quartiles and their characteristics. Within each quartile, the railroads' bonds had very similar yields. In the second quartile, for example, yields ranged from 4.1% to 4.3%, indicating that the market regarded these securities as similar. We then look within each quartile and identify any railroads whose median bond rating

³⁰ We do not take up the question of the accuracy of the ratings here, but instead focus on the ratings' effect on markets. Hickman (1958), however, uses longer-run data to show that ratings had strong predictive power for subsequent defaults.

³¹ We use the full distribution of all traded bonds, not just railroad bonds, in constructing these yield quartiles.

TABLE 3
RAILROAD BOND YIELD QUANTILES

Prerating Yield Quartile	Minimum Yield	Maximum Yield	Mean Yield	Percent Aaa	Percent Aa	Percent A	Percent Baa or Lower	Mean Rating (1 = Aaa)	Median Rating
1	.029	.041	.039	.841	.028	.000	.131	1.421	Aaa
2	.041	.043	.042	.298	.602	.101	.000	1.769	Aa
3	.044	.047	.045	.255	.292	.453	.000	2.188	A
4	.048	.076	.053	.000	.052	.652	.297	3.511	A

NOTE.—Distribution of average yields in the period prior to ratings of bonds of different railroads, sorted into quartiles, and median ratings those railroads' bonds received in each yield quartile. Railroads that received a rating below the median for their quartile are designated as having received a negative surprise. The presented mean yield, distribution of ratings, and mean rating within each yield quartile are computed based on bond-week observations that match the sample used in our prior regression analysis in table 4.

was higher or lower than the median bond rating within the quartile. We regard these as cases in which the rating assigned by Moody was likely to have been a surprise or contained some new information. We then test whether railroads that received negative surprises in their ratings saw their yields change relative to other railroads with the same rating following the introduction of the ratings.

The data in table 3 indicate that there were many surprise ratings. For example, whereas 84% of the bond-week observations in our sample in the lowest yield quartile ("Quartile 1") received the highest median firm-level rating (Aaa), 3% received a rating of Aa, and 13% received a rating of Baa or lower. The median rating in that quartile was Aaa, so the latter two ratings were negative surprises. We note, however, that many of the railroads receiving positive surprises in our sample had relatively low yields, consistent with a median rating of Aa, and they instead received a rating of Aaa. The low yields of these bonds indicate that they were already perceived to be extremely unlikely to default, and we would expect the effect of good news regarding the riskiness of those bonds to be fairly small. By contrast, it is possible that bonds receiving negative surprises may have been more sensitive to news about their credit risk. Many of these negative surprises were received by railroads whose bonds were already perceived as riskier (i.e., in higher-yield quartiles), and the more negative assessments by Moody likely conveyed that these investments were more "speculative."³² In addition, many of the negative surprises in the first quartile were several notches below the median rating. We also observe relatively fewer positive firm surprises in our data. We therefore focus our analysis on the effects of negative ratings surprises in our empirical analysis, relative to a control group of railroads receiving either no surprise or a positive surprise. In appendix C, we present alternative specifications in which we separately test for the effects of positive surprises and for large surprises.

We study the effect of these negative firm surprises in the context of a model with bond and week fixed effects, as well as controls for rating levels interacted with trends. This structure has the added benefit that it not only tests for effects of plausibly measured surprises relative to market expectations, but it also implicitly tests for convergence in yields within ratings levels.³³ Our initial estimating equation will therefore be

³² We would expect bond prices to be especially sensitive to news when they are close to the default versus nondefault region. Yet whether the negative surprises we identify in our sample were more informative about default and repayment than our positive surprises is ultimately an empirical question.

³³ Developments in the market for corporate debt that were ongoing at the time when ratings were introduced make it especially important to control for ratings levels interacted with trends. Figure A.9 illustrates the time path of the mean yields of bonds from April 1908 to April 1910 by the level of ratings they were eventually given. The yields of bonds at each rating level were clearly declining over the year prior to the introduction of the ratings, and the declines

$$y_{ijt} = \alpha_i + \gamma_t + \delta_1 \text{negsurprise}_{jt} \times \text{postRatings}_t + \sum_n \pi_n \text{RatingLevel}_{ni} \times \text{trend}_t + \epsilon_{it}, \quad (1)$$

where y_{ijt} is the yield to maturity of bond i issued by railroad j in week t ; α_i and γ_t are bond and week fixed effects; negsurprise_{jt} is an indicator for whether or not railroad j 's rating was worse than the median rating of their yield quartile and was therefore a negative surprise; postRatings_t is an indicator equal to 1 for all weeks following the introduction of Moody's ratings, which occurred on 23 April 1909; RatingLevel_{ni} are indicators for the railroad rating level assigned by Moody; and trend_t is a time trend measured as weeks since the start of the sample. Our key parameter of interest is δ . We cluster standard errors at the bond level.

If ratings conveyed information, we would expect the yields of railroads receiving a negative surprise (i.e., the treated railroads) to increase relative to those issuers that received no surprise or a positive surprise, after the 1909 manual was released.

B. Bond Market Response

1. Information Provision

We begin by comparing the week-by-week evolution in differential yields. Figure 2 plots the differences in yields between railroads that received a negative surprise and those that did not over the 2 years of our data, as estimated from a regression like the one specified in equation (1), where δ is allowed to vary every week and all estimates are relative to the month when ratings were released (which is therefore the omitted group not shown in the figure).³⁴ The figure presents a clear indication that rating surprises did indeed change bonds' yields. Importantly, the difference between the average yield for the bonds of railroads that received a negative surprise in their rating and those that did not (conditional on the rating level they would eventually receive) was stable over time and hovered around zero for the year prior to the introduction of the ratings. This suggests that there were no differential preexisting trends between the two groups,

were greater for bonds that received lower ratings ex post. This reflects a general pattern of declining credit spreads in the market for corporate debt during the recovery from the Panic of 1907 and the related economic downturn. Any empirical analysis that simply tests for differences in yield levels or credit spreads following the introduction of ratings would confound any effect of the ratings with the ongoing trend of declining yields.

³⁴ The regression used to produce the estimates in the figure is like eq. (1), except rather than estimating an average postratings effect of a negative surprise, a series of negative surprise $\times$ date interactions are included, to estimate the difference-in-differences relative to the excluded date—the month of the introduction of ratings, April 1909. The interactions between the ratings levels and time trends are also included, as are the bond fixed effects.

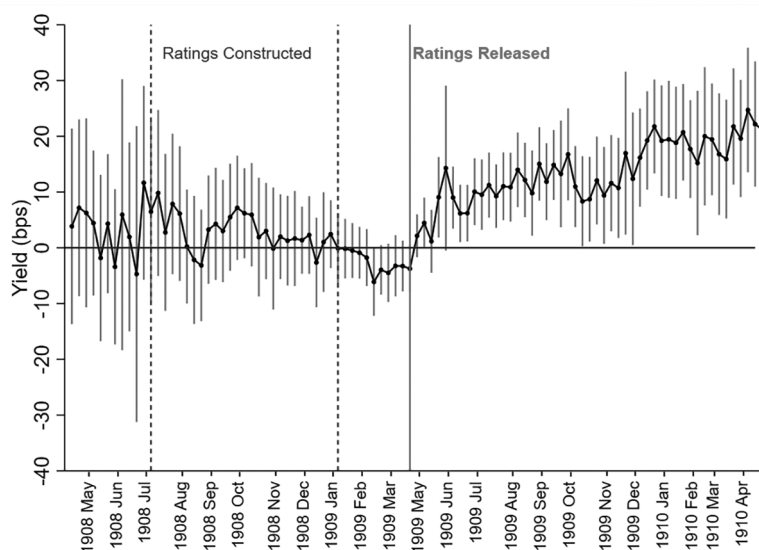


FIG. 2.—Ratings surprises and bond yields. The figure presents the differences in yields between bonds whose ratings constituted a negative surprise and other rated bonds—those that either received a positive surprise or no surprise. The differences plotted are estimated from regressions of yield-to-maturity on indicators for a negative surprise interacted with indicators for each week, with the week prior to the introduction of ratings excluded. The regressions also include bond fixed effects, as well as rating-level fixed effects interacted with time trends (see text). The figure also includes lines representing 95% confidence intervals. The solid vertical line indicates the release of ratings; the dashed lines denote the time period in which Moody constructed the ratings.

conditional on our controls. Yet immediately after the introduction of the ratings, the difference begins to increase, with the railroads that received a negative surprise commanding higher yields. This differential increase in yields stabilizes at about 6 months after the publication of the ratings, with the yields on bonds whose ratings was a negative surprise being about 20 basis points higher relative to those of other rated bonds.

Notably, the effect displayed in the figure is not one of an instantaneous, discreet jump but rather a gradual increase that starts just after ratings are introduced. This is consistent with the fact that it may have taken some time for Moody's volume to reach some institutions and traders (April 23 is the publication date of the volume, not necessarily the date when it was received) and that the significance of the ratings and the analysis underpinning the ratings may not have been understood immediately. Instead, the volume gradually became more influential over time, and its contents' influence over bond yields increased accordingly. Still, the flat nature of the relative yield difference prior to ratings and the subsequent sharp change in the slope of the difference after ratings

were introduced using relatively high frequency data support a causal interpretation of the effect of ratings surprises on bond yields.

Figure 2 also helps rule out the possibility that Moody was simply good at predicting future changes in yields and that he rated bonds according to his expectations of these price changes. While the volume was published April 23, 1909, most of the information it contained and upon which the ratings were based was as of the end of the railroads' last fiscal year, which ended June 30, 1908. For some railroads, Moody included a brief description of changes in conditions that had occurred since the close of the fiscal year, which were likely added after the ratings tables had been created. These often stated that no events of significance had occurred, and even when some were described, the very latest date of the additional information was January 1, 1909. Since the ratings were computed well before April, the timing of the increase in yields following this date strongly points to a causal effect of the publication of the ratings on the market.³⁵ By contrast, the period during which the ratings were constructed but not yet released (depicted with dashed lines) reveals no such response.

Next, we formalize the analysis from the figure in table 4, where we present results from estimating variations of equation (1). In column 1, we start by regressing the yield to maturity on the interaction of a dummy variable equal to 1 if a firm received a negative surprise interacted with an indicator for the postratings period. Given that it took about 6 months for the effect of the ratings to be fully captured in yields, we present a "donut" specification, where the postratings indicator is equal to 1 only for the second half of the postratings year to estimate the effects. The regression also includes bond and week fixed effects and rating fixed effects interacted with time trends. Consistent with the picture presented in figure 2, we observe a 16 bps increase after 26 weeks in the yield of bonds among firms with negative ratings surprises.

These findings hold across a range of specifications, including utilizing alternative measures of yields, postratings windows, and alternative measures of surprises. Focusing solely on the transactions that occur in the 6–12 months after ratings were released throws out a significant fraction of the observations, however, and can potentially affect estimates and standard errors. We next use a more flexible framework to incorporate the entire postratings period. Specifically, in column 2, we interact the negative surprise indicator with a time trend and then also include a time trend interacted with a postratings indicator. This specification allows for the

³⁵ In addition, the results are nearly identical if we compute surprises using only yields prior to January of 1909 rather than April of 1909 (not shown). This is reassuring because it helps rule out concerns that any predictive power of the ratings between January and April can influence our measure of surprises and their effects on yields.

TABLE 4
EFFECT OF RATINGS SURPRISES ON YIELDS

Yield (basis points)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Negative surprise $\times$ post	15.50*** (4.87)						
Negative surprise $\times$ trend (weeks) $\times$ post		.47*** (.19)					
Negative surprise $\times$ trend (weeks)		-.04 (.16)					
Negative surprise $\times$ weeks since			.41*** (.12)	.47*** (.12)	.43*** (.15)	.32*** (.16)	.35*** (.14)
Implied 12-month ATE (bps)	16	24	21	25	22	16	18
95% CI, bps	[6,25]	[5,44]	[9,34]	[12,37]	[7,38]	[5,33]	[4,33]
Bond fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects $\times$ trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm $\times$ trend fixed effects							
Maturity $\times$ trend fixed effects				Yes		Yes	Yes
Level of surprise	Firm	Firm	Firm	Firm	Bond	Bond	Bond
Postperiod duration, weeks	26-52	52	52	52	52	52	52
R ²	.908	.905	.905	.908	.906	.917	.920
Observations	11,423	15,478	15,478	15,478	15,220	15,220	15,220

NOTE.—Effects of “surprises” (i.e. deviations in ratings from the median for those in the same yield quartile based on their mean yield among all traded bonds prior to the introduction of ratings) on secondary market bond yields trading on the NYSE. Column 1 presents a “donut” specification; we regress yield to maturity in basis points on a dummy variable equal to 1 for a negative surprise interacted with a dummy variable equal to 1 if the bond transaction occurs 26 weeks after Moody’s securities ratings are released in April of 1909 (post). This specification also includes bond and week fixed effects and rating fixed effects interacted with time trends. The preperiod includes the year prior to the release of ratings, while the post period includes the period 6–12 months following their disclosure. Implied 12-month average treatment effects (ATEs) for the primary coefficient of interest as well as 95% confidence intervals (CIs) for those 12-month estimates are included below the specification. In col. 2, we interact negative surprise with a time trend and then also with a time trend interacted with an indicator variable for the postratings period. Unlike the donut of col. 1, it includes all 12 months following release of ratings in the post period. In col. 3, negative surprise is interacted with a variable that equals 0 prior to ratings being released and then after is the weeks since they were released. Column 4 is the same as col. 3 but also includes 10 bond-level maturity group fixed effects interacted with a time trend. Column 5 is the same as col. 3, but the surprises are calculated at the bond level, rather than the firm level. Column 6 is the same as col. 5 but also includes firm $\times$ trend fixed effects, so that only the within-firm variation is used to estimate the effect of the surprises. Column 7 is the same as col. 6 but also includes 10 bond-level maturity group fixed effects interacted with a time trend. Standard errors clustered at the bond level are in parentheses.

** Significant at 5%.

*** Significant at 1%.

effects of negative surprises to trend over time, instead of simply focusing on the average effect in the post period, and also estimates any preexisting differential trends in yields prior to the release of the ratings. Reassuringly, the results indicate that there was no ongoing differential trend in the yields of bonds that received a negative surprise but that a differential trend emerged following the publication of ratings.

In column 3, we estimate a more parsimonious version of the specification of column 2 and interact our negative surprise variable only with a variable that measures the weeks since the ratings were released, capturing the postratings differential trend. Yields differentially increased for firms that received a negative surprise; the estimates imply a 21 bps increase over 12 months. This is our preferred specification, in the sense that it achieves a good balance between parsimony and power. In column 4, we show that these results are quite similar, even controlling for 10 bond-level maturity group fixed effects interacted with time trends, suggesting that our estimated effects are unlikely to be an artifact of differential trends among bonds of differing maturities.

A potential concern regarding these estimates could be that the firms that received negative ratings surprises also differed along some other dimensions we may only imperfectly observe. If those firms happened to have differential exposure to some unanticipated economic event that coincided with the publication of the ratings, its impacts could potentially confound our estimated effects of ratings. Although we believe this is unlikely, we cannot dismiss that possibility completely. To help address this concern, we exploit variation in ratings surprises among bonds issued by the same firm.

We first recalculate the surprises without aggregating by firm. These bond-level surprises are more likely to be measured with error, due to the thinness of trading of many bonds; the noisy preratings average yields of individual bonds taken from just a few observations distort the comparisons made to construct the surprises, and thus we interpret their effects with some caution.³⁶ Yet they vary within firms, since the bonds of a given railroad often differed in terms of their market yields as well as the ratings they received. Thus, within the same firm, some bonds may have received ratings that were worse than expected given their prerating yields, while other bonds did not. We use this source of variation in column 5 by re-estimating our preferred specification from column 3 but include an indicator for a negative surprise at the bond level, rather than at the firm level. The results of this specification are very similar to the estimate reported in column 3 and help address concerns that our findings are influenced by the way we aggregate the data to estimate firm-level surprises.

³⁶ This also results in the loss of a very small number of observations (<2%) with insufficient liquidity prior to ratings to estimate a surprise at all.

In column 6, we investigate whether bonds that received negative surprise ratings saw differential changes in their yields after ratings were released, relative to other bonds of the same firm. In this specification, we include firm fixed effects interacted with time trends and therefore remove any potential time-varying confounds at the firm level. Again supporting a causal interpretation of our findings, the estimates indicate differential patterns in bond yields following rating surprises consistent with our prior estimates. In column 7, we show that results are essentially unchanged when we control for a major driver of bond-level yields, maturity, utilizing the same set of maturity controls from column 4.

The results thus far imply that the introduction of securities ratings did indeed change bond market yields. This is strong evidence that even in the absence of financial regulations based on ratings, and even in the presence of abundant information on bonds and bond issuers, letter-graded ratings can provide information about credit risk that has a meaningful impact on market prices. In table A.2, we show that these results are robust to alternative specifications in which we control for the prerating yield quartiles (interacted with trends), rather than ratings (interacted with trends). That is, whereas our main specifications compare firms with similar ratings whose bonds had different yields, these alternative specifications compare firms whose bonds had the same yields but received different ratings. Using this alternative source of variation, we obtain very similar results.³⁷

Another more subtle concern could be that ratings predicted news events, and market yields responded to those news events rather than the ratings. We think this is unlikely since markets responded in just the first few weeks after ratings were released. In section B of the appendix, we show a significant impact of surprises on yields even when we restrict the analysis to the 4 or 8 weeks after the publication of the manual. In addition, these effects are largely unchanged when we control flexibly for all news we could find about these firms during that period (or even removing firms with any news at all). See appendix section B for more details.

It also important to note that our findings are always in relative terms, since new information about some bonds may affect the willingness of market participants to transact in other bonds. While we cast our findings as evidence that bonds with negative surprises saw their yields rise relative to those that do not, it is equally valid to say that those that did not receive a negative surprise (because theirs was positive or there was no surprise) saw their yields fall relative to those that did. In section C of the appendix, we also show explicitly that the bonds of firms with positive surprises saw their yields fall (relative to those with negative surprises or no surprise at

³⁷ Our calculations of yields to maturity are somewhat imprecise, and in table A.2, we also show that when we reestimate our regressions using perpetuity yields (coupons divided by prices), rather than yields to maturity, our results become much stronger.

all). In general, we fail to reject any difference in magnitude of effect size for positive versus negative surprises, except among large surprises, where we find some evidence consistent with negative bond-level surprise effects being more substantial. We discuss these results in detail in appendix section C.

Next, we further explore our main findings. Though the analysis of bond-level surprises in table 4 helps address concerns that firm characteristics may be biasing our effects, one may still worry about these potential biases for our main analysis based on railroad-level surprises. To address this issue, we study the robustness of the effects of firm-level surprises to the inclusion of key railroad characteristics that were determinants of the ratings. To ease comparison, column 1 of table 5 simply replicates column 3 of table 4, our preferred specification based on weeks since. Column 2 controls in addition for our proxy for market-based expectations for a bond's risk (the prerating yield) interacted with weeks since the ratings were released and obtains very similar results. This helps dispel concerns that surprises may be simply picking up differential trends in yields for railroads in the same yield quartile with different prerating yield levels.

Another possibility is that our measures of surprises are picking up other bond information that Moody's utilized to construct ratings. As we show in table 2, of all the features that Moody's explicitly took into account in his ratings tables, factor of safety was the most important predictor of ratings. In column 3 of table 5, we add factor of safety, interacted with weeks since the ratings release. As one would expect, we find that railroads with a higher average factor of safety saw less of an increase in yields in the post period. Importantly, the effect of negative surprises is virtually unchanged. In column 4, we show that the estimated coefficient on negative surprise is also robust to controlling for average income and interest per mile. It is also possible that bonds more or less sensitive to interest rate risk—a factor not explicitly taken into account by Moody—may have seen yields change differentially. Yet our results are robust to controlling for the average duration of the railroad's bonds (see col. 5). It is important to note that the variables we included thus far are on their own very strong predictors of the actual ratings assigned by Moody, which in turn are one of the two components we use to estimate surprises. In column 6, we control for both the bonds' prerating average yield, the other key piece of information we use to estimate surprises, and a predictor of ratings used by Moody. Remarkably, including this measure, as well as the average prerating bid-ask spread as a measure of liquidity, makes barely any dent on the effect of negative surprises. These results are important because they suggest that credit ratings may affect markets beyond the information that would be contained in linear assessments of a bond's risk, either those formed from relevant firm or bond characteristics or from market expectations (i.e., yields).

These findings are also not the result of ratings simply conveying soft or insider information about specific firms. In column 7, we use the same set of prerating observable control variables times *Weeks since* from column 6 to instrument for *Weeks since* times *Negative surprise* in a 2SLS regression. Consistent with these data explaining variation in surprises, even after all fixed effects, the regression has a reasonably strong first stage: the surprises are partly predictable using information that was publicly available and generally accessible. In addition, the instrumented surprises have a statistically significant effect on postratings yields that is similar in magnitude to the ordinary least squares (OLS) estimate reported in column 6. This suggests that factors other than observable financial data, such as soft or insider information, could not have been entirely responsible for the observed market effects of the ratings surprises.

Since firm-level surprises are largely predictable with public information, in which other way, then, can ratings convey information? One possibility is that the coarse nature of credit ratings, assigned to groups comprising substantial sets of securities, may be a more effective way to provide information to market participants, who would otherwise have to process complex information. In this sense, our results may be consistent with theoretical work that shows that in some contexts coarse signals can be more illuminating than precise ones, because they provide more clarity for interpreting the intended message sent (e.g., Martel et al. 2022), and with experimental work showing that simplifying complex information helps consumers make better financial decisions (e.g., Carpenter et al. 2021). Alternatively, the introduction of letter-graded ratings may have caused some investors to categorize particular bonds differently and consider issues with the same rating, rather than issues that shared other attributes, as comparable (e.g., Bordalo, Gennaioli, and Shleifer 2013; Ellis and Matslioglu 2022). Although we cannot distinguish among these different potential mechanisms, our findings suggest that in the historical context of the first-ever ratings, what may have mattered was not only the level of information disclosure but the manner in which this information was actually packaged, simplified, and presented to the market. Market participants were already familiar with ordinal ratings systems from experience with commercial credit reporting, which likely contributed to the influence of securities ratings (Carruthers 2022, 160–64).

2. Variation in Impact of Ratings

The results presented in tables 4 and 5 are remarkable for their consistency. Even when controlling for firm fixed effects, or for key determinants of the main components of the surprises themselves, the estimated effects of negative rating surprises are, while modest in magnitude, quite robust. They suggest that, even in an environment in which market participants

TABLE 5
EFFECT OF SURPRISES, CONTROLLING FOR RATING PREDICTORS

	OLS						2SLS
Yield (basis points)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Weeks since ×							
Negative surprise	.41*** (.12)	.41*** (.12)	.34*** (.13)	.34*** (.13)	.34*** (.14)	.33*** (.14)	.64*** (.23)
Prerating yield		4.35 (11.01)				-6.51 (13.31)	
Factor of safety			-.006 (.005)	-.006 (.005)	-.006 (.005)	-.007 (.006)	
Average income				.000004 (.00002)	.000002 (.00002)	.000001 (.00002)	
Interest per mile				-.00004 (.0001)	-.00002 (.0001)	-.00002 (.0001)	
Duration					-.004 (.018)	-.006 (.012)	
Prerating bid-ask						5.70 (12.98)	
Implied 12-month ATE (bps)	21	21	18	18	18	17	33

95% CI, bps	[9,34]	[9,34]	[4,32]	[4,31]	[4,32]	[3,31]	[10,56]
Bond fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects $\times$ trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F -statistic							14.94
R^2	.905	.905	.905	.905	.905	.905	.008
Observations	15,478	15,478	15,478	15,478	15,478	15,478	15,478

NOTE.—Effects of “surprises” (i.e., deviations in ratings from the median for those in the same yield quartile based on their mean yield among all traded bonds prior to the introduction of ratings) on secondary market bond yields trading on the NYSE after controlling for other potential observable confounds. Column 1 regresses yield to maturity in basis points on a dummy variable equal to 1 for a negative surprise interacted with a variable that equals 0 prior to ratings being released and then after are the weeks since they were released. This specification also includes bond fixed effects, week fixed effects, and rating fixed effects interacted with weeks since ratings were released. The preperiod includes the year prior to the release of ratings, while the postperiod includes the 12 months following their disclosure. Column 2 is the same as col. 1 but also interacts weeks since with prerating mean yield-to-maturity (prerating yield). Column 3 is the same as col. 1 but also interacts weeks since with a firm’s average bonds’ factor of safety as a control variable. Column 4 is the same as col. 3 but also interacts weeks since with a firm’s bonds’ average earnings available to pay interest (average income) and interest per mile as additional control variables. Column 5 is the same as col. 4 but also interacts weeks since with a firm’s average bonds’ duration as an additional control variable. Column 6 is the same as col. 5 but also interacts weeks since with a firm’s bonds’ average preratings release bid-ask spread percent and yield-to-maturity as additional control variables. Column 7 uses all control variables in col. 6 but instead as IVs within a 2SLS regression where weeks since $\times$ negative surprise is the endogenous variable and yields are again the outcome variable of interest. In this case, CIs are weak instrument robust confidence sets based on the two-step identification robust 95% confidence sets proposed by Andrews (2018) based on linear combination tests, as implemented in the twostepweakiv package in Stata using 1,000 grid points. See also Andrews, Stock, and Sun (2019). All characteristics are computed at the firm level by taking the mean for each bond-week observation in our sample, as in table 2. Standard errors clustered at the bond level are in parentheses.

*** Significant at 1%.

have strong incentives to acquire information, ratings can still improve information provision. A natural question then arises: Are there characteristics of firms where information provision via ratings is more or less impactful?

Negative surprises had an impact on yields when they updated market expectations downward. This finding suggests, however, that those surprises may have had less of an impact on firms with strong reputations. We study this possibility by analyzing whether connections to prominent financiers served as a substitute for credit ratings. As shown by Frydman and Hilt (2017), in the early twentieth century, top underwriters utilized their positions on railroad boards to monitor those firms, thereby alleviating their financial constraints and helping them grow. These tight relationships with bankers were observable by market participants—firms often advertised their directors and, importantly, the lists of board members were also published in investor manuals. In column 1 of table 6, we therefore allow the effect of negative ratings surprises to vary by whether the railroad had many connections to the top underwriting firms of the time through their board of directors—specifically, that they were among the top quartile in such connections, with at least three elite financiers among their directors. We find that the negative effect of surprises on yields was essentially undone for the most connected firms. Next, we compare the effects of negative surprises on the most connected railroads (in col. 2) with those on moderately connected railroads (in col. 3) and with those that had no connections to top underwriting firms (in col. 4). Firms with no connections saw large increase in yields following a negative surprise, while the effect was much smaller and insignificant for those railroads with a high number of top underwriters on their boards.

These results suggest the role of ratings does appear to have been mediated by a firm's reputation, such as the signal imparted by the quality of a their underwriters. If this reputation-enhancing mechanism helped resolve problems related to asymmetric information, then the fact that ratings surprises only impacted firms without the mechanism in place suggests the effects of ratings were also related to asymmetric information. We explore this connection in more detail below.

C. Interpretation of Effects of Ratings on Yields

We interpret the estimated effects of ratings on yields (table 4) as evidence that securities ratings can be an effective tool for information provision. By making complex data easier to evaluate, ratings refined investors' expectations regarding the likelihood of defaults, which was then reflected in changes in those bonds' yields.

An alternative explanation of these effects might be that the ratings were inaccurate yet nonetheless induced changes in yields because investors

simply believed them and sold off the bonds that received negative surprises. If this were the case, the ratings could have produced the observed impacts on yields without actually improving investors' understanding of the risks of securities; ratings may have served merely as a focal point that led to commonality in beliefs (e.g., Boot, Milbourn, and Schmeits 2006; Morris and Shin 2007).

This interpretation would imply that ratings had powerful effects that increased the impact of the information being provided (i.e., it would still support our definition of information provision), although not through improving the equilibrium level of fundamental information in prices. Yet the lack of reversion in prices (fig. 2) suggests this is unlikely to be the most plausible explanation for the effects we observe. A full year after the release of ratings, which should have been a sufficiently long period for the inaccuracy of ratings to have been revealed, there was no indication of a weakening of the effects of negative surprises, much less a reversal.

TABLE 6
HETEROGENEITY IN EFFECT OF RATINGS SURPRISES ON YIELDS

Yield (basis points)	All (1)	Many Bankers on Board (2)	Some Bankers on Board (3)	No Bankers on Board (4)
Weeks since × Negative surprise	.54*** (.13)	.011 (.31)	.45*** (.17)	.61*** (.15)
Negative surprise × many top underwriters on board	-.63** (.26)			
All interactions	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	Yes
Rating fixed effects × trend (weeks)	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
Underwriter measure (top 10 bankers on board)	≥3	≥3	[1,2]	0
R ²	.906	.921	.900	.910
Observations	15,478	3,745	7,748	3,985

NOTE.—Heterogeneity in effects of “surprises” (i.e., deviations in ratings from the median for those in the same yield quartile based on their mean yield among all traded bonds prior to the introduction of ratings) on secondary market bond yields trading on the NYSE. Column 1 regresses yield to maturity in basis points on a dummy variable equal to 1 for a negative surprise interacted with a variable that equals 0 prior to ratings being released and then after are the weeks since they were released (weeks since) and interacted with a dummy variable equal to 1 if the firm is in the top quartile of the distribution of connections to elite underwriters, with at least three on their board (many top underwriters on board). This specification also includes bond fixed effects, week fixed effects, and rating fixed effects interacted with weeks since ratings were released. The preperiod includes the year prior to the release of ratings, while the postperiod includes the 12 months following their disclosure. Columns 2–4 estimate the effect of the interaction of negative surprise and weeks since, only among the subset of firms where either more than 2, 1 or 2, or no top underwriters are on the board, respectively. Standard errors clustered at the bond level are in parentheses.

** Significant at 5%.

*** Significant at 1%.

Further evidence against this interpretation comes from the response to ratings from investment bankers. The acceptance of inaccurate ratings by investors should have created profitable trading opportunities for well-informed actors. Yet the best-informed and most sophisticated investors of the time, investment bankers, strongly opposed Moody's innovation of providing ratings (Stimpson and Mahoney 2008). The response to the introduction of ratings was "in no circles . . . more hostile than among investment bankers," and their hostility was specifically attributed to the ratings' tendency to "narrow the price spread between trading points" and influence "the resale of bonds" as well as the "original sale of new issues" (Harold 1938, 16). By making ordinary investors better informed, ratings likely reduced the informational advantage of investment bankers and other sophisticated investors.³⁸ The bitterness with which investment bankers greeted the ratings is evidence supporting our preferred interpretation of their effects.

VI. Ratings and Functioning of Financial Markets

We next analyze the equilibrium response of the functioning of financial markets to the introduction of the ratings. If Moody's innovation did indeed help small investors to become better informed, or otherwise reduced the informational advantage of sophisticated investors, it may have resolved problems related to asymmetric information and improved market liquidity. This effect would not have been produced by the particular ratings given to bonds, and whether they constituted surprises, but simply by the fact that bonds were given ratings. We therefore need a different empirical design than the one utilized above, focused on whether bonds were rated, rather than the specific ratings bonds received.

A. *Empirical Method and Predictions*

Moody's volume of ratings included the majority of large railroad systems and some smaller ones, too, but it excluded a relatively small number of NYSE-listed railroad bonds. Though these unrated bonds constitute a natural control group for the analysis of the effect of the ratings on liquidity, Moody's choices over which railroads to leave out were not random. For example, rated bonds tended to be substantially more liquid (e.g., smaller bid-ask spreads) than unrated bonds, though there was substantial overlap in the distribution of spreads across both groups (table A.3). To evaluate the effect of the presence of ratings on market liquidity, we need to

³⁸ When Moody discussed his books with traders, he was informed by an "old Wall Street buccaneer . . . if you begin to flaunt too many facts, there won't be much inside knowledge left to work on; you will be spoiling our game" (Moody 1933, 91).

examine plausibly exogenous variation in Moody's propensity to rate particular railroads' bonds.

Moody's ratings included all of the railroads with liquid, high-quality (low-yield) bonds, which were of the greatest interest to investors. However, some of the railroads rated by Moody had some relatively lower-quality, less-liquid issues outstanding—issues that were similar to the bonds that were not rated. Many of these small, riskier bond issues were originally the obligations of smaller railroads that had been acquired by a larger system and were secured by collateral that was less valuable (per mile) than other bonds of the same system.

When Moody rated the bonds of a railroad, he rated all of its bonds, not merely the more liquid or safe issues.³⁹ This meant that lower-quality bonds of railroads that also had high-quality bonds outstanding received ratings simply because they were liabilities of a railroad with other high-quality bonds outstanding. By contrast, similar lower-quality bonds issued by railroads that had no high-quality issues were less likely to be rated. We therefore use the average yields of the other outstanding bonds of the same issuer as an instrument for whether or not a bond was rated. If the average yield of the other bonds of the same issuer did indeed influence Moody's decision to rate a particular bond but did not cause any changes to its bid-ask spread in the period after the ratings were introduced, then it represents a valid instrument for whether or not a bond was rated.

As with our analysis of the effects of ratings on yields, we account for any ongoing trends in the differences between rated and unrated bonds using a linear time trend. We estimate the following regression via 2SLS:

$$\begin{aligned} liquidity_{it} = & \alpha_i + \gamma_t + \theta_1 rated_{it} \times postRatings_t + \theta_2 rated_{it} \\ & \times trend_t + \beta X_{it} + \epsilon_{it}, \end{aligned} \quad (2)$$

where $liquidity_{it}$ is proxy for issue i 's liquidity (such as bid-ask spread) in week t ; α_i and γ_t are bond and week fixed effects; $rated_{it}$ takes a value of one for issues that were rated by Moody's for the weeks after the ratings' release; $postRatings_t$ is an indicator equal to 1 for all weeks following the introduction of Moody's ratings, which occurred on April 23, 1909; $trend_t$ is a time trend; and X_{it} includes various characteristics of issue i , such as its average yield and bid-ask spread during the period prior to the introduction of the ratings, interacted with a postratings indicator. Since $rated_{it}$ appears twice in (2), our specification contains two endogenous regressors. We therefore use two instruments: the average yield of the other outstanding issues of the same railroad as issue i interacted with the

³⁹ As noted above, the analysis underpinning the ratings required Moody to create a seniority ranking of all outstanding bonds for the railroads he rated and calculate the available income for each. It was therefore quite natural to rate all bonds and the marginal cost of doing so was likely low.

postratings period ($\bar{y}_{-i} \times \text{postRatings}_i$) and the average yield of other outstanding issues interacted with a trend ($\bar{y}_{-i} \times \text{trend}_i$). The main parameter of interest is θ_1 . Our key identifying assumption is that comparing observationally equivalent bonds of different issuers, the bond belonging to an issuer with other bonds with lower yields is more likely to have been rated. We exploit that increase in likelihood of being rated to study whether receiving a rating has a causal impact on the bonds' market liquidity.

B. Effects of Getting Rated—IV Results

In table 7, we present our estimates of the effects of being rated on bid-ask spreads. As a baseline in column 1, we present the results from an OLS regression of bid-ask spreads on a dummy variable equal to 1 if a bond was rated, after controlling for issue fixed effects, time fixed effects, the prerating mean bid-ask spread for that issue interacted with a post-ratings dummy, and the prerating mean yield for that issue interacted with a postratings dummy. We also allow for differential trends by rated status. The estimated value of the parameter θ_1 from equation (2) indicates that rated bonds saw their spreads fall by 60 bps in the weeks following the introduction of ratings. That said, it is important to reiterate the caveat that new information about some bonds may also affect market participants' views (and their trading) for other (in this case, not rated) bonds. Thus, it is possible that the market perceived a lack of ratings as information about relative asymmetric information, and so therefore one could equally interpret our results as indicating that the introduction of ratings caused a rise in bid-ask spreads for unrated bonds relative to rated ones.

While the OLS estimates are suggestive of an effect of ratings on liquidity, as we noted previously, they are subject to selection concerns. Column 2 of table 7 presents the same equation estimated via 2SLS, to alleviate bias caused by selection effects arising from Moody's choice of which bonds to rate, while the corresponding first stages (with matching column numbers) in their entirety are shown in table A.4. Consistent with our expectations, the corresponding first-stage regressions in column 2 of table A.4 show that the estimated parameter on the instrument is large and negative, indicating that bonds issued by railroads whose other bonds had lower yields were much more likely to be rated. Specifically, we find that a 1 percentage point increase in the preratings yields of the other bonds of a railroad reduced the likelihood that a bond was rated by about 11%, after controlling for the risk and liquidity of that bond prior to the ratings' release.

The parameters in column 2 of table 7 indicate that the 2SLS estimate of the effect of ratings on bid-ask spreads is considerably larger than OLS

TABLE 7
IV RESULTS: EFFECT OF RATINGS ON BID-ASK SPREADS

	OLS	2SLS		
	(1)	(2)	(3)	(4)
Post × rated issue	−.006* (.003)	−.031*** (.011)	−.029*** (.010)	−.026*** (.010)
Trend × rated issue	.0001 (.0004)	.002 (.001)	.002 (.001)	.002 (.002)
Weak IV CI, post × rated		[−.052, −.010]	[−.049, −.010]	[−.044, −.008]
Bond fixed effects	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
Week fixed effects × prerating yield	Yes	Yes	Yes	Yes
Week fixed effects × prerating spread	Yes	Yes	Yes	Yes
Preperiod illiquidity (%)			≥60	≥80
Kleibergen-Paap F-statistic		16.4	28.2	60.7
Observations	4,991	4,991	2,036	1,044
Observations not rated (%)	6	6	10	15
Number of bonds	346	346	166	88
Number of bonds not rated	25	25	19	14

NOTE.—Estimates of the effect of ratings on bid-ask spreads, using OLS and the IV approach of eq. (2), estimated via 2SLS. The regressions are estimated using 12 weeks of data before and after the introduction of ratings, for a total of 24 weeks of data. The weak instrument robust confidence sets are two-step identification robust 95% confidence sets proposed by Andrews (2018), based on linear combination tests, as implemented in the twostepweakiv package in Stata using 1,000 grid points. See also Andrews, Stock, and Sun (2019). The prerating yield and the prerating spread are calculated as the mean yield and spread for each issue using all the observations 12 weeks before treatment. Column 1 is an OLS regression of a bond’s bid-ask spread on a dummy variable equal to 1 if an issue (bond) is rated interacted with a dummy variable equal to 1 after ratings are released. It also includes bond, week, week × prerating yield, and week × prerating spread fixed effects. Column 2 is the same as col. 1 but uses the IV approach from eq. (2). In particular, after controlling for an issue’s yield prior to the release of ratings, we use the yields of other bonds that are part of that same firm in the same period, as an instrument for being rated (because of Moody’s propensity to rate all bonds of a given firm). The same-issuer prerating yield is calculated using observations over the 12 weeks before the introduction of ratings. Column 3 is the same as col. 2 but restricts the analysis to only those issues with a mean bid-ask spread at or above the 60th percentile in the period prior to ratings being released. Column 4 is the same as col. 3 but for those above the 80th percentile. The corresponding first stages of the regressions in cols. 2–4 are shown in table A4. Robust standard errors, clustered by bond, are in parentheses.

* Significant at 10%.

*** Significant at 1%.

and large in absolute terms—310 bps. This is in fact larger than the mean value of the preratings spread for rated bonds. It is important to note, however, that the local average treatment effect (LATE) is obtained from relatively illiquid bonds that were rated only because they were issued by railroads that had some high-quality issues. In fact, our first-stage coefficients start to increase significantly once we restrict the sample to only those bonds that were highly illiquid (above the 60th percentile in the

bid-ask spreads distribution in the prerating period).⁴⁰ This suggests that the instrument primarily operates through these highly illiquid bonds. While these estimated treatment effects are substantial, they are actually plausible within this subgroup.

To illustrate this point more clearly in columns 3 and 4 of table 7, we reestimate our regressions focusing on only those issues with bid-ask spreads above the 60th and above the 80th percentiles of the preratings distribution, respectively.⁴¹ Despite having higher levels of initial spreads, we find very similar percentage point declines, implying much smaller percent declines than in column 2. In particular, these equate to about a 1.6 standard deviation decline in spreads among those more illiquid bonds. At the same time, the first-stage *F*-statistic rises substantially. This is because, just as we noted above, the LATE is the average treatment effect for compliers, and the compliers in our data are likely to be highly illiquid securities. Very liquid, frequently traded securities were rated no matter the characteristics of the other outstanding bonds of the railroad. By contrast, illiquid securities, with less market interest, would have been marginal bonds and may have been rated or not, depending on whether the firm had other, more prominent securities.⁴²

These results are consistent with work by Kelly and Ljungqvist (2012), who show in modern data that proxies for increases in information asymmetry driven by reduced equity analyst coverage widen bid-ask spreads. The effect of ratings in our sample was likely quite small for the high-quality issues that were the focus of Moody's volume and of investors' interest, which had very low bid-ask spreads before the introduction of the ratings. But for the less liquid, higher-yielding issues that were rated for somewhat arbitrary reasons and had initially very wide spreads, the effect of the ratings was quite substantial. In summary, then, what this evidence suggests is that among those securities most likely to benefit from improved liquidity (i.e., highly illiquid securities), ratings were actually able to tighten their bid-ask spreads substantially. These findings also support the analysis in section V, since they suggest that incremental information made public through ratings disclosures narrowed bid-asks spreads, mitigating problems related to asymmetric information.

⁴⁰ We use the full distribution of all quoted bonds, not merely railroad bonds, to calculate these liquidity cutoffs.

⁴¹ In addition, restricting the sample to illiquid bonds helps us improve the comparability of rated and nonrated bonds, which are much fewer in our data, since Moody included the majority of railroad systems in his manual. In the bid-ask sample, only 7.2% of the bonds and 6% of the observations were not rated. When we focus on the bonds above the 60th percentile of illiquidity, e.g., those numbers are 10% and 11.4%, respectively.

⁴² We address concerns regarding test size distortions caused by weak instruments by including weak-instrument robust confidence intervals, based on Andrews (2018) and discussed in Andrews, Stock, and Sun (2019). These exclude 0 for all IV specifications.

TABLE 8
REDUCED-FORM ESTIMATES

	Reduced Form			Falsification	
	(1)	(2)	(3)	(4)	(5)
Post $\times$ other bonds yields	.353*** (.111)	.394*** (.121)	.385*** (.131)	.080 (.252)	.255 (.212)
Trend $\times$ other bonds yields	-.019 (.016)	-.026 (.020)	-.025 (.023)	-.010 (.014)	-.004 (.008)
Bond fixed effects	Yes	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes	Yes
Week fixed effects $\times$ prerating yield	Yes	Yes	Yes	Yes	Yes
Week fixed effects $\times$ prerating spread	Yes	Yes	Yes	Yes	Yes
Preperiod illiquidity (%)		≥ 60	≥ 80	≤ 20	≤ 60
Kleibergen-Paap F -statistic, first stage	16.4	28.2	60.7	1.0	.5
Observations	4,991	2,036	1,044	1,015	3,010
Observations not rated (%)	6	10	15	1	2
Number of bonds	346	166	88	57	183
Number of bonds not rated	25	19	14	2	6

NOTE.—Estimates of reduced-form relationship between instrument used in estimating eq. (2), yield on other bonds of the same railroads, and bid-ask spreads. The regressions are estimated using 12 weeks of data before and after the introduction of ratings, for a total of 24 weeks of data. The prerating yield and the prerating spread are calculated as the mean yield and spread for each issue using all the observations 12 weeks before treatment. Column 1 is a regression of bid-ask spreads on the instrument, interacted with a postratings indicator and with a time trend. It also includes bond, week, week $\times$ prerating yield, and week $\times$ prerating spread fixed effects. Column 2 is the same as col. 1 but restricts the analysis to only those issues with a mean bid-ask spread at or above the 60th percentile in the period prior to ratings being released. Column 3 is the same as col. 2 but for those above the 80th percentile instead of the 60th. In col. 4, we present the results of a falsification test in which we restrict the sample to the most *liquid* bonds—those in the 20th percentile of prerating spreads or below. In col. 5, we present another falsification test, with the sample restricted to bonds in the 40th percentile of spreads or below. Robust standard errors, clustered by bond, are in parentheses.

*** Significant at 1%.

A potential source of concern regarding our instrumental variable (IV) estimates could be that they are driven by declining bid-ask spreads among the bonds of large railroad systems.⁴³ Since those systems were most likely to have a high-quality (low-yield) bonds outstanding, any change in their bonds' bid-ask spreads would be conflated with the effects of being rated in our framework. In table 8, we show that this is unlikely to be responsible for our results. In the table, we present the reduced-form estimates of our IV, with columns 1–3 displaying the estimates from the samples of columns 2–4 in table 7. As expected, the estimates are positive and statistically significant—the lower the yields of the other bonds of the same railroad,

⁴³ An additional concern is that the differences in the distribution of liquidity prior to the introduction of between-rated and unrated bonds may affect our estimates. Yet our 2SLS coefficients are robust to restricting the sample to only those bonds in the overlapping part of the spreads distributions (results not shown).

the lower the bid-ask spread in the postratings period, which we ascribe to the greater likelihood of getting rated. Yet in columns 4 and 5, when we restrict the sample to the most liquid bonds, we find no effect. As can be seen clearly in the table, almost all very liquid bonds were rated (only 2 out of 57 in col. 4 and only 6 of 183 in col. 5 were not rated). For these bonds, whether other bonds in the same system were of interest to investors and had low yields was immaterial.⁴⁴ These falsification tests show that the instrument does not predict declines in bid-ask spreads for liquid bonds and suggest that our main findings are not simply an artifact of comparing bonds in prominent railroad systems with those of small railroad systems. Only the most illiquid bonds of the rated systems saw their bid-ask spreads decline, which is inconsistent with the notion that changes in the spreads of all the bonds of large systems are driving our results.

Bid-ask spreads are, of course, not the only measure associated with liquidity and a well-functioning financial market. In table 9, we use the same IV approach to study the effects of ratings on the block size of trades (the number of shares traded when a bond trades).⁴⁵ We find that even though being rated does not seem to change the probability that a bond will trade on a given day among our compliers (col. 1), the number (col. 2) and probability (col. 3) of single-lot trades increased (conditional on any trades for that security occurring). As the par value of most bonds was \$1,000, at a time when nominal GDP per capita was less than \$500, even a single-lot trade was an immense amount of money. To the extent that small investors participated in this market, it would likely have been reflected in single-lot trades. Single-lot trades were the most common transaction type in our data (~21%), and although not all of these trades were initiated by small investors, virtually all trades by small investors would have been in single lots. While not as strong or clear as our results on bid-ask spreads, these findings could suggest that ratings encouraged more trading in illiquid securities (our compliers) among retail investors. By contrast, we find no evidence of such a rise among large trades (col. 4), again suggesting that if ratings attracted increased trading interest for these illiquid securities it was probably most concentrated among retail traders. Taken together, these results support the notion that ratings narrowed bid-ask spreads in bond markets and may have also opened up the trading of more illiquid securities to small investors.

⁴⁴ This finding is consistent with the lack of a valid first stage in these samples. However, it is essential for us to obtain our main IV results to include the most illiquid bonds in our sample, for which the first-stage and reduced-form estimates start to rise.

⁴⁵ These data were compiled from the NYSE's daily reporting of bond transactions, which lists the block size of every trade. We collect this data for all bond trades for every Wednesday over the same interval of time for which we collected bid-ask data. See fig. A.4 for a partial example of intraday transactions for a single day.

TABLE 9
IV RESULTS: EFFECT OF RATINGS ON BLOCK SIZE

	Two-Stage Least Squares			
	Any Trade (1)	Number of 1-Lot Trades (2)	$\mathbf{1}_{1\text{LotTrade}}$ (3)	Number of Large Trades ($\geq 75\text{th}$ Percentile) (4)
Post April 23, 1909 $\times$ rated issue	.099 (.183)	.574** (.280)	.428** (.200)	-.104 (.708)
Trend $\times$ rated issue	-.003 (.011)	-.028 (.021)	-.021 (.014)	.031 (.071)
Weak IV robust confidence set, post $\times$ rated	[.034, 1.113] [.042, .814]			
Bond fixed effects	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
Week fixed effects $\times$ prerating yield	Yes	Yes	Yes	Yes
Week fixed effects $\times$ prerating spread	Yes	Yes	Yes	Yes
Kleibergen-Paap F -statistic	7.9	10.7	10.7	10.7
Dependent variable mean	.277	.403	.333	.663
Observations	8,975	2,418	2,418	2,418
Number of bonds	359	235	235	235
Number of bonds not rated	68	52	52	52

NOTE.—Estimates of effect of ratings on trade size, and in particular on number of lots (of \$1,000 par-value bonds) per transaction. All regressions are versions of eq. (2) estimated via 2SLS, with different second-stage outcomes as shown in the column headings. We use the yields of other bonds that are part of that same firm in the same period as an IV for being rated (because of Moody's propensity to rate all bonds of a given firm). All regressions include bond, week, week $\times$ prerating yield, and week $\times$ prerating spread fixed effects. The prerating yield and the prerating spread are calculated as the mean yield and spread for each issue using all the observations 12 weeks before treatment. Similarly, the instrument, the same-issuer prerating yield, is calculated using observations over the 12 weeks before the introduction of ratings. The regressions are estimated using 12 weeks of data before and after the introduction of ratings, for a total of 24 weeks of data. The weak instrument robust confidence sets are two-step identification robust 95% confidence sets proposed by Andrews (2018) based on linear combination tests, as implemented in the twostepweakiv package in Stata using 1,000 grid points. See also Andrews, Stock, and Sun (2019). Column 1 is a 2SLS regression of the probability of any transaction occurring that day for that bond on an instrumented endogenous dummy variable equal to 1 if an issue (bond) is rated interacted with a dummy variable equal to 1 after ratings are released. Column 2 is the same col. 1, but the dependent variable is the number of standard (single) lot trades in a day, conditional on at least one transaction occurring that day for that bond. Column 3 is the same col. 1, but the dependent variable is a dummy variable equal to 1 if at least one standard (single) lot trade occurs in a day, conditional on at least one transaction occurring that day for that bond. Column 4 is the same col. 1, but the dependent variable is number of large trades in a day, defined as those in the 75th percentile or above, conditional on at least one transaction occurring that day for that bond. The 75th percentile is a 16-lot trade. Robust standard errors, clustered by bond, are in parentheses.

** Significant at 5%.

VII. Conclusion: Ratings, Bond Markets, and Information

This paper investigates the effects of the introduction of the first-ever bond ratings on financial markets. In 1909, John Moody produced and sold to investors a volume containing letter-graded ratings of the majority of railroad bonds listed on the New York Stock Exchange. Utilizing a variety of newly collected data, our analysis shows that this financial innovation improved the liquidity of corporate bonds and, when the ratings conveyed negative information relative to investor expectations, caused a modest but appreciable increase in their yields relative to other bonds. In the years since 1909, bond markets and the institutions associated with them have evolved considerably. What insights do these results offer for credit ratings and bond markets generally?

First, our unique historical setting allows us to provide credible evidence that ratings can help transmit information on credit risk to bond markets. A vast empirical literature has generally cast any effects of ratings on prices and firms as evidence of information transmission. Yet because ratings are embedded into modern financial regulations and investment mandates, their effects may be completely unrelated to information. A rating that brings a security closer to or further away from the threshold of "investment grade," for example, will change expected demand for that security and therefore its price. The fact that Moody's 1909 ratings had significant effects at a time when there were no regulations or mandates tied to ratings demonstrates that ratings can actually influence the functioning of capital markets through information provision.

Our findings also offer some suggestive evidence on the mechanisms through which ratings may transmit information. Like modern ratings, Moody's 1909 ratings were largely explainable with public information that was broadly accessible. Their effects suggest that simplifying complex and multidimensional information by partitioning securities into coarse letter-graded categories may in fact be effective means of information transmission and that the form in which information is presented may matter. Our results on trade sizes suggest that the effects of ratings were driven in part by greater participation among small investors and institutions, for whom acquiring and processing information was likely costly.

Although our historical setting allows us to isolate the information effects of ratings, it is unclear whether the magnitudes we estimate from 1909 are relevant to modern financial markets. This leads to our third insight, which is that reliance on ratings may actually influence information production outside rating agencies. Before the introduction of credit ratings, sophisticated investors devoted considerable resources to analyzing the risks of bonds. Once ratings were introduced, investors began to rely on them quite heavily (Harold 1938), and the incorporation of ratings into financial regulations only increased their influence. The centrality of

modern credit ratings has reduced the incentives of sophisticated modern investors to analyze the creditworthiness of relatively safe securities, other than during financial crises (Johnston, Markov, and Ramnath 2009; Hanson and Sunderam 2013; Dang, Gorton, and Holmström 2020). Modern financial institutions effectively delegate information production for relatively safe securities to ratings agencies. Despite tremendous technological advances that have improved the availability of data and the processing of information over the past century, the reliance on ratings has diminished the incentives of investors to use those technologies and analyze the risks of relatively safe securities. As a result, credit ratings today may in fact have significant information-based effects, as they did in 1909.

A fourth and closely related point is that our analysis documents the origins of the role of ratings in creating safe assets. Safe assets are vital for the functioning of the economy, and in modern financial markets, privately created Aaa-rated assets account for a large share of the supply (Gorton, Lewellen, and Metrick 2012; Gorton 2017). Our results focusing on ratings surprises show that Moody's 1909 volume led to convergence in bond yields within ratings levels, as investors began to view bonds of similar ratings as having similar credit risks. Of particular significance was Moody's designation of some bonds as Aaa, his highest rating. Moody stated that those issues "are dependent for their prices on the current rates for money, rather than the fluctuations in earning power" (Moody 1909, 193), a characterization that is consistent with modern definitions of information-insensitive securities (e.g., Dang, Gorton, and Holmström 2020). Following the publication of Moody's volume, corporate bonds rated Aaa quickly became important in the investment decisions of commercial banks and other institutional investors.⁴⁶ Over the twentieth century, the composition of safe assets produced by the financial system changed in ways that made ratings particularly important.⁴⁷ Our results suggest that when they were first introduced, ratings immediately began to shape investors' assessments of the riskiness of securities. Thus, third-party risk assessments that are credible to investors can help create safe assets, even when these assessments have no bearing on financial regulations that restrict the types of assets that institutions can hold.

Finally, in the wake of the 2008 financial crisis, serious concerns were raised about the quality of credit ratings (Benmelech and Dlugosz 2009,

⁴⁶ Harold (1938, 20) describes a commercial banker, known as "Triple-A James," who insisted on buying only Aaa bonds, noting that he was "typical of thousands in the banking profession." Harold also mentions the importance of highly rated bonds to other types of institutional investors, e.g., insurance companies and trust companies.

⁴⁷ Especially after 1980, moneylike debt instruments and mortgage-backed securities became increasingly important safe assets (Gorton, Lewellen, and Metrick 2012); a rating of AAA/Aaa is crucial to their status as safe (Dang, Gorton, and Holmström 2020).

2010). Our results show that when implemented in their original form—as an evaluation of securities sold to investors, with no role in regulations and no perverse incentives created by the modern issuer-pays business model—ratings increased participation and improved the functioning of financial markets. Credit ratings were an important financial innovation, one of many that over the past century have improved transparency and information access in financial markets, potentially improving the ability of firms to raise financing and the allocation of capital in the economy more broadly and thereby contributing to long-run economic growth.

Data Availability

Code replicating the tables and figures in this article can be found in Bernstein, Frydman, and Hilt (2025) in the Harvard Dataverse, <https://doi.org/10.7910/DVN/B0VD17>.

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